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Summary

Senior Machine Learning Engineer with nearly **4.5 years** of experience in machine learning, specializing in large-scale systems for enterprise solutions. Architected and spearheaded the implementation of a **AIOps platform** and production-grade **Mixture of Experts** models focusing deep domain specialization. Seeking to leverage expertise in **LLMs** and scalable machine learning architectures to drive impact in cutting-edge technology projects.

Technical Skills

Programming: Python, C++, Javascript

Frameworks: PyTorch, Transformers, CrewAI, LangGraph, DSPy, Langfuse

Generative AI: LLMs, RAG, Mixture of Experts, Prompt Engineering, Reasoning Models

Computer Vision: Object Detection, Image Processing, OCR

NLP: Transformers, Language Models, NER, Intelligent Document Processing

MLOps & Deployment: Docker, REST APIs (FastAPI framework), MLFlow, RabbitMQ

Experience

Senior Software Engineer - ML and Agentic AI

August 2024 – Present

RANDSTAND DIGITAL

Remote

DocuSign

August 2025 – Present

- Architected and deployed production-scale **AI agent systems** for DocuSign's sales outreach teams, implementing **CrewAI**-based multi-agent workflows for automated lead generation and personalized email campaigns, serving enterprise sales operations.
- Engineered end-to-end data infrastructure supporting AI operations, developing **dbt** transformation pipelines and **Airflow DAGs** for orchestrating complex data workflows across **Snowflake** and production databases.
- Built and deployed **PII anonymization** microservices using **Docker** containerization and **AWS ECS** task deployment, ensuring compliance requirements for sensitive customer data in ML pipelines.

CISCO

August 2024 – August 2025

- Architected and led implementation of large-scale **AIOps platform** providing enhanced Assurance capabilities across diverse Cisco products (**ThousandEyes**, **Meraki**, **Catalyst Center**) for enterprise customers.
- Engineered and deployed production-grade **Mixture of Experts (MoE)** models, developing novel modular framework to convert standard **LLMs** into specialized MoE architectures, serving internal enterprise teams.
- Spearheaded development of **Network Root Cause Analysis (RCA)** solution by generating synthetic reasoning datasets and training bespoke models; successfully deployed for enterprise-wide internal testing.

Freelance Consultant - AI/ML

August 2023 – August 2024

FREELANCE

Noida, India

- Implemented end-to-end **NL2SQL** pipeline with **Human-in-the-loop** verification, labelling, training and validation loop with best **MLOps** principles for a **Fortune 500** company.
- Developed automated **Video Tagging** pipeline leveraging **Computer Vision** to identify and tag fashion items from product catalogs within videos.
- Built **AI Teaching Assistant** agent using **RAG** and multimodal data processing (docs, videos, web) for automated generation of teaching materials and student evaluations.
- Engineered multi-lingual **ID document parsing** system (passports, govt. IDs) extracting key entities from various file types.
- Developed **Computer Vision** systems for vehicle detection and tracking in driving school analysis.
- Architected advanced **RAG** applications incorporating query transformation and re-ranking, utilizing **Vector Databases (ChromaDB, Qdrant)**.
- Fine-tuned **LLMs (LLaMa-2, Mistral 7B)** using **PEFT, QLoRa**, and applied advanced prompt engineering (**CoT, ReAct**).
- Utilized **Quantization, Distillation, and Pruning** in models to significantly enhance deployment efficiency in limited-resource settings.

Machine Learning Engineer

August 2021 – August 2023

ALTHEA.AI (previously Wisteli Informatics)

Noida, India

- Led full lifecycle of **5** key Proofs-of-Concept, translating requirements into production-ready ML solutions for document intelligence and healthcare domain.
- Developed and deployed **multi-modal models** for automated classification, layout detection and data extraction (**LayoutLM**) from complex documents (medical reports, invoices, KYC data).

- Implemented custom model + heuristics pipeline achieving **78%** accuracy on challenging borderless table extraction tasks.
- Improved overall pipeline accuracy by **20%** through strategic application of **YOLOv5** and advanced **OpenCV** image processing techniques.
- Built and deployed ML models for insurance **fraud/abuse detection**.
- Engineered and deployed scalable **REST APIs (Flask)** on **AWS EC2** using **Docker** and **RabbitMQ** for asynchronous processing.
- Developed specialized solutions including **Tuberculosis detection** pipeline (X-ray analysis) and image denoising engine (**U-Net** based autoencoder) improving **OCR** quality.
- Implemented advanced image forgery detection techniques (**RANSAC, Autoencoders, Noise Print**).
- Managed team of **4** developers and mentored **7** interns, driving project execution and fostering technical growth.

Education

Babu Banarasi Das University <i>Bachelor of Technology, Computer Science & Engineering – CGPA: 8.7</i>	Lucknow, UP, India <i>2021</i>
Dr. Rizvi Learners Academy <i>Intermediate (10+2)</i>	Jaunpur, UP, India <i>2016</i>
Dr. Rizvi Learners Academy <i>High School</i>	Jaunpur, UP, India <i>2014</i>